Module 13

Technical Workouts

1. Understand ML Engineering Fundamentals

a. Machine Learning vs ML Engineering
b. Model development vs model serving
c. Production ML systems
d. End-to-end ML workflows
e. ML product lifecycle

2. Understand API Fundamentals

a. What is an API
b. REST architecture
c. Request-response lifecycle
d. API endpoints
e. API design principles

3. Understand what FastAPI is and its architecture.

a. Learn how to set up a FastAPI project and run the development server.
b. Create API routes using different HTTP methods.
c. Understand request handling, response generation, and data validation using Pydantic.
d. Explore FastAPI's automatic API documentation (Swagger UI)

4. Understand Model Serving

a. Loading trained models
b. Prediction workflows
c. Input validation
d. Output formatting
e. Inference pipelines

5. Understand ML API Development

a. Building prediction endpoints
b. Handling user inputs
c. Returning predictions
d. Error handling
e. Scalable API architecture

6. Understand API Security

a. Authentication awareness
b. Authorization concepts
c. Rate limiting awareness
d. Input sanitization
e. Secure ML APIs

7. Understand API Documentation

a. API specifications
b. Endpoint documentation
c. Request examples
d. Response examples
e. Developer usability

8. Understand Production Considerations

a. Latency awareness
b. Scalability
c. Reliability
d. Monitoring
e. Maintainability